

Summary Report

Predicting Fast-Growing Firms Using Bisnode Data

1. Introduction

Identifying firms with high future growth potential is an important objective for investors, lenders, and economic policy makers. This report presents predictive models that estimate the probability that a firm will become a fast-growing company. They are designed to support decision-making by providing both probability predictions and classification rules that balance business risks.

2. Data and Target Definition

We start from the raw Bisnode firm-level panel dataset and restrict the sample to the period 2010–2015, in line with the assignment requirements and to ensure consistent observation of firm dynamics over time. Variables with extremely high missingness (>80%) are removed, as they provide limited usable information and would require heavy imputation. Firm identifiers are also cleaned to guarantee consistency across years.

Since growth is inherently a dynamic concept, we construct a complete firm–year panel. This means that firms with missing observations in certain years are explicitly included, with missing financial values recorded as such. This step preserves the correct time structure of the data and allows us to compute forward-looking growth measures in a consistent way.

Sales are converted into millions and transformed using logarithms to reduce skewness and improve interpretability. Economically implausible values, such as negative sales, are removed to avoid distortions in size and growth calculations.

Defining Fast Growth

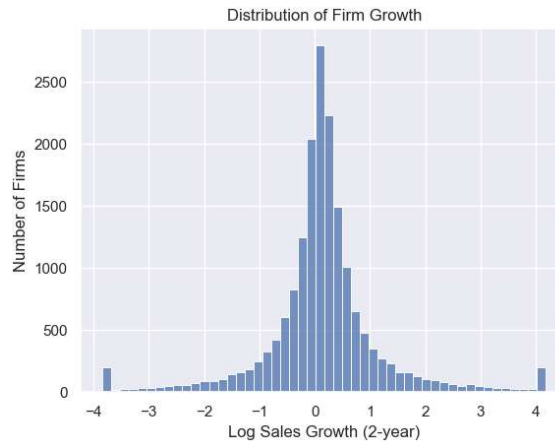
A key modelling decision was defining “fast growth,” with two alternatives considered: one-year growth (2013 vs 2012) and two-year growth (2014 vs 2012). A two-year growth horizon was selected and defined as:

$$growth = \log(sales_{2014}) - \log(sales_{2012})$$

This approach was chosen for three main reasons:

- a) Economic relevance: Corporate finance literature shows that sustainable growth is better captured over multi-year horizons rather than single-year changes, which can be affected by temporary shocks.
- b) Noise reduction: Two-year growth smooths short-term volatility in sales and provides a more stable measure of firm expansion.
- c) Operational usefulness: The horizon remains short enough to allow early identification of promising firms.

Figure below shows the distribution of 2-year firm growth. Most firms experience relatively small changes in performance, with growth rates concentrated around zero. However, a small group of firms achieves exceptionally high growth, creating long tails in the distribution.



To capture firms that significantly outperform their peers, fast-growing firms are therefore defined as those belonging to the **top 20%** of the growth distribution.

3. Feature Construction and Exploratory Analysis

To improve predictive performance, additional variables were created to capture firm characteristics, financial condition, and managerial factors commonly associated with firm growth. Industry classifications were simplified into broader groups to reduce noise and improve model stability, while also enabling comparison between manufacturing and service firms. Geographic and urbanization indicators were retained to control for regional economic differences.

Firm age was calculated from founding year and complemented with a squared term to capture non-linear growth patterns, as younger firms often grow faster before stabilizing. Managerial characteristics, including CEO age, ownership structure, and gender indicators, were included to reflect differences in leadership and governance.

Financial variables were expressed as ratios to ensure comparability across firms of different sizes and to capture liquidity, profitability, and capital structure. Extreme values were trimmed to reduce the influence of accounting outliers. Missing data were handled using median imputation combined with missing-value indicators, allowing the models to retain observations while preserving potential information contained in missingness. The final modelling dataset contains **17,927 firms and 73 engineered variables**.

Lowess smoothing plots further revealed several important relationships:

- Firm size shows a strong negative relationship with fast growth probability, indicating that smaller firms are more likely to experience rapid expansion.
- Profitability demonstrates a non-linear relationship with growth, where firms with very low profitability rarely grow quickly, while moderate profitability is associated with higher growth potential.
- Liquidity shows a mild positive association with growth, suggesting that firms with greater financial flexibility may be better positioned to expand.
- CEO age displays a declining relationship with growth probability, indicating that younger leadership may be associated with more aggressive expansion strategies.

Based on these findings, a final predictor set consisting of **30 variables** was selected. We aimed to capture both economic fundamentals and organizational factors influencing firm growth while avoiding excessive model complexity.

4. Model Development and Selection

To predict fast-growing firms, three complementary modelling approaches were developed:

- 1) Logistic Regression (Logit): serves as a transparent baseline and allows direct interpretation of how firm characteristics affect growth probability.
- 2) Random Forest: captures complex interactions and non-linear relationships between predictors without requiring strict functional assumptions
- 3) Gradient Boosting: sequentially improves prediction performance by focusing on observations that are difficult to classify and is often effective in structured tabular datasets.

All models were trained using the same set of 30 predictors to ensure comparability. Model hyperparameters were optimized using cross-validation to reduce overfitting and ensure stable out-of-sample performance. First, we used the Area Under the Receiver Operating Characteristic Curve (AUC), which measures how effectively a model distinguishes fast-growing firms from other firms.

Model	AUC
Gradient Boosting	0.777
Random Forest	0.772
Logit	0.766

Gradient Boosting achieved the highest predictive accuracy, although differences across models were relatively small. This suggests that the selected predictors contain strong growth information that can be captured by both linear and non-linear models.

5. Classification Approach and Business Loss Function

In this study, *missing a fast-growing firm was considered significantly more costly than incorrectly identifying a firm as fast-growing*. From a strategic perspective, failing to identify high-growth firms may lead to missed investment or expansion opportunities. To reflect this, the following loss function was applied:

- False Positive Cost = 1
- False Negative Cost = 5

For each model, classification thresholds were optimized using **five-fold cross-validation** to minimize expected business loss rather than relying on default probability cutoffs. This approach ensures that model evaluation directly reflects economic decision-making. Table below summarizes the final comparison across models:

Model	AUC	Threshold	Expected Loss	Predictors
Gradient Boosting	0.777	0.170	1878.8	30
Random Forest	0.772	0.162	1902.2	30
Logit	0.766	0.186	1930.8	30

Gradient Boosting achieved the lowest expected loss and therefore delivered the strongest overall business performance. While Random Forest produced similar predictive accuracy, it resulted in

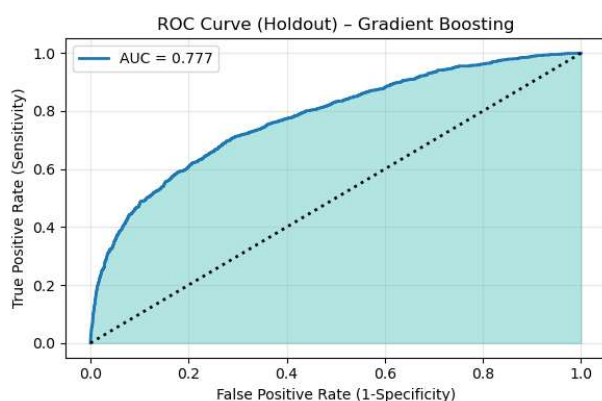
slightly higher classification costs. Logistic regression showed the lowest predictive performance but remains useful as an interpretable benchmark.

The optimal classification threshold for Gradient Boosting is approximately 0.17, reflecting the asymmetric cost structure that prioritizes identifying fast-growing firms. Considering both predictive accuracy and economic impact, Gradient Boosting was selected as the preferred model for further evaluation and sector-specific analysis.

6. Model Validation and Final Performance Evaluation

We further evaluated the selected model using an independent holdout sample to confirm that it performs well on unseen firms and not only within the training data. The GB model was trained on 70% of firms and tested on the remaining 30%, using the previously selected optimal threshold of 0.17. The model maintained strong predictive performance with a holdout AUC of 0.777, which indicates good ability to distinguish between fast-growing and non-fast-growing firms.

The confusion matrix provides additional insight into classification performance. The model correctly identifies 768 fast-growing firms while missing 308, and correctly classifies 3,008 firms as non-fast-growing while incorrectly flagging 1,295 firms as fast-growing. This outcome reflects the intended trade-off implied by the loss function, where missing high-growth firms is considered more costly than generating false alerts, and therefore the model is designed to prioritize detection of potential high-growth opportunities.



Confusion Matrix (Holdout) – Gradient Boosting

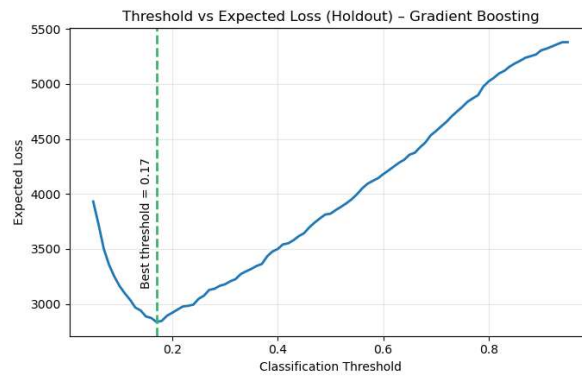
Actual class	Predicted class	
	No fast growth	Fast growth
No fast growth –	TN 3008 55.9%	FP 1295 24.1%
Fast growth –	FN 308 5.7%	TP 768 14.3%

These results translate into the following performance measures:

- Accuracy: 70.2%
- Precision: 37.2%
- Recall: 71.4%

The relatively high recall indicates that the model successfully identifies a large share of fast-growing firms, which aligns with the business goal of minimizing missed growth opportunities. Although precision is lower, this trade-off is expected because missing a high-growth firm is assumed to be significantly more costly than incorrectly flagging a non-growing firm.

The graph below shows how expected loss varies across thresholds. The results indicate that the chosen threshold of 0.17 minimizes expected loss by balancing false positives and false negatives. Lower thresholds would generate too many false alerts, while higher thresholds would increase the risk of missing fast-growing firms.



Overall, the holdout validation confirms that the Gradient Boosting model provides reliable predictions and supports business decision-making by identifying firms with strong growth potential while controlling classification costs.

7. Sector Comparison: Manufacturing vs Services

To evaluate whether model performance differs across industries, the selected Gradient Boosting model was applied separately to manufacturing and services firms using the same loss function and cross-validation approach. The results show that predictive performance varies across sectors.

	Group	N	AUC	Expected loss	Best threshold
0	manufacturing	45	0.630	8.2	0.098
1	services	200	0.584	34.6	0.308

The model performs better for manufacturing firms, achieving an average AUC of 0.63 and a lower expected loss of 8.2. In contrast, performance is weaker for services firms, with an AUC of 0.58 and a higher expected loss of 34.6.

The optimal classification threshold also differs substantially between sectors. Manufacturing firms require a lower threshold (approximately 0.10), indicating that growth signals are stronger and easier to detect. Services firms require a higher threshold (approximately 0.31), suggesting that growth patterns in this sector are more heterogeneous and harder to predict. These results highlight that industry structure influences predictability and that sector-specific decision rules may improve targeting of fast-growing firms.

8. Conclusion

To sum up, this analysis demonstrates that firm growth can be predicted with reasonable accuracy using a combination of financial, firm-level and managerial information. By defining fast growth over a two-year horizon and explicitly incorporating business costs into the classification decision, we move beyond purely statistical performance and focus on economic relevance. Among the models considered, Gradient Boosting delivered the best balance between predictive accuracy and expected classification cost, making it the preferred choice for identifying high-growth firms. The results also highlight meaningful differences across industries, suggesting that growth dynamics and prediction difficulty vary between manufacturing and services. Taken together, the findings provide a practical and data-driven tool that can support early identification of high-potential firms and improve strategic investment or policy decision-making.

* The full code, data preparation steps, and modelling implementation used in this analysis are available in the GitHub repository: <https://github.com/alinaiman/Data-Analysis-3-assignment-2->